PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Definitions and measurement of health literacy in health and medicine research: a systematic review
AUTHORS	Urstad, Kristin; Andersen, Marit; Larsen, Marie; Borge, Christine Ra°heim; Helseth, Sølvi; Wahl, A

VERSION 1 – REVIEW

REVIEWER	Kristiansen, Maria University of Copenhagen, Department of Public Health & Center for Healthy Aging
REVIEW RETURNED	11-Sep-2021

GENERAL COMMENTS	Thank you for the opportunity to review this important and well-written manuscript reporting on a systematic review into how health literacy is defined and measured in current health literacy research. The manuscript represents an important contribution to the booming field of health literacy research, and I have only minor suggestions for improvements. The Introduction is comprehensive and argues well for the need for this piece of work with relevant references cited. Overall, the Methods are described in sufficient detail. However, a bit more explanation of the more unusual approach of including articles published in two randomly selected months would be helpful. This is discussed further in the Discussion but a bit more insights earlier in the manuscript would be helpful. At page 7, the text reads: "The search yielded a total number of 1031 citations. Endnote™ Version X9 was used to manage the generated research articles. After removing duplicates, 951 citations remained. All records not published for the first time in 2019 were removed, leaving 414 articles". To avoid confusion, I would suggest adding "for screening" to the sentence. The Results section is supported by relevant figures and tables. Page 10, lines 9-24: it would be helpful for the reader to learn a bit about what types of public health settings were involved in the 60 articles, what populations were engaged and what (quantitative) methods the articles report upon. That is, a bit more description of the study characteristics in the text. Table 1: I would suggest reversing the order of the reporting here, starting with category 1 rather than 2.

	The Results section is quite brief, relying too much on tables (1-3) that are not sufficiently explained and summarized in the text. Please elaborate on the findings reported in the tables to help the reader capture the key findings. Please consider arranging articles in tables 1-3 in alphabetical order. The Discussion and Conclusion are comprehensive, and raises important reflections pertaining to results and methodological strengths and weaknesses of the work conducted. Would it be possible to raise more concrete implications of your work, that is, how do we move forward in order to overcome the identified inconsistency between between health literacy definitions and instruments in future health literacy research? Furthermore, the review included only quantitative measurements. What might be the value of including qualitative methods in order to create more meaningful approaches to this field?

REVIEWER	Ng, Jia Hwei Donald and Barbara Zucker School of Medicine at Hofstra/Northwell, Medicine
REVIEW RETURNED	06-Oct-2021

GENERAL COMMENTS	Review response for - How health literacy is defined and measured in current health and medicine research The authors Urstad et al have asked an important research question in the field of health literacy □ do researchers use HL instruments that match the chosen HL definition? We commend the authors for embarking on this research project. However, the data presented in this paper is sparse. Additional detail is needed. The discussion section is presented in a vague manner, creating more questions rather than providing clarity. I have provided some recommendation below on how to improve further. Purpose and Argument • The purpose and background is clear in the introduction. • The authors introduced the concept of “established health literacy theory” only in the discussion. This needs to be mentioned in the introduction section. Methods • As mentioned by the authors, the study design is limited by the time frame. However, the number of articles is likely sufficient to answer the research question. [May need additional review by epidemiologists/ statistician who have expertise in sampling for systematic review]. • Did the investigators measure inter-rater reliability between the reviewers? If so, please report the data. Data/ Results section • The data presented is sparse. The lack of granularity makes it difficult to identify next steps for future research. • It is insufficient to just say that 60 articles have clinical health focus and 60 in public health settings. The authors could categorize the articles by sectors/ subspecialty etc. • Page 10, line 49. Section on Health literacy definitions and instruments.

	o In order to keep the paper consistent, please move paragraph 1 in this section to be paragraph 2 (The paragraph on health literacy definition should come before HL definition). • Certain sections of the data are important and helps show the extent of the problem (Tables 2 and 3). However, it would be even more effective to provide some granularity on which group of articles that have mutual levels or confliction levels of HL definition and HL instruments. That way, the readers can identify targets of intervention. Based on Appendix 2 (coding for data extraction), the investigators have collected several other variables including: perspective, country of origin, population health status, type of disease, type of HL instrument (generic/ disease-specific). Thus, the investigators should make full use of these data. Discussion section: • Overall, the discussion section needs polishing and rethinking. Although it is well-organized, the arguments/ recommendations are often vague and abstract. Several of the paragraphs seems to create more questions than help clarify. Further details are necessary. To put it in another way, the authors often "tell", but do not "show". The following are some examples: • Page 18, line 17. "As a result, there is a risk of missing important information [specifically, what is missing? What do you consider important?] about health literacy considered important in the studies'…" • Page 18, line 22. "This concern [what specific problem? Please give more details] was previously addressed by health literacy researchers" [23]. The article in reference 23 talks about many different aspects of health literacy. I suggest choosing a specific aspect that is most relevant to your research question. • Page 18, lines 23-34. "The current study … is the first review to investigate… from the perspective of an established health literacy theory". o This is the first time the authors have mentioned this. In the introduction section, the authors have referenced Nutbeam and Sorensen. The way it is written, it read as if they are experts in the field, not that Nutbeam have established a health literacy theory. Since the audience of BMJ open are typically wide, I recommend that the authors include several sentences or one short paragraph describing more about the established health literacy theory. • Page 18, lines 50-52. "Despite the increased development of questionnaires with a broader comprehensive focus [somewhat confusing – broad vs focus – which one?], it might take more time for these instruments [which instruments? – It seems like the authors are talking about new questionnaires being develop. But they could also be talking about the new vital sign.] to become established in the health literacy research field." • Page 19, lines 3-6. " …, most of the focus has been on measuring individual health literacy, but all sides of the health literacy equation (including health systems and health professionals) and social processes are important to research…". o Organizational health literacy and health literacy related to health professionals are viewed in a different context. In particular, organization health literacy is measured by separate instruments. The statement above is not really relevant to the study question. I recommend removing it. • Page 19, lines 10-12. "Definitions should be the basis of measurement. However, measurement should continually inform definitions".

	o Vague concept. Unclear what the sentences actually mean. Need to clarify further/ given more concrete examples. • Page 19, lines 52-55. “An interesting finding is that, among the eight studies…. were in the context of mental health.” o This is the first time this study finding is mentioned. Please include in the result section. This is a good example of some granular data that is needed in the results section.

VERSION 1 – AUTHOR RESPONSE

Comment from author:
Correction: When revising the manuscript, we discovered that one of the articles was referred to twice in Table 3. This is now revised. Correct information is that 79 out of the 120 included articles provided a health literacy definition (not 80).
Reviewer: 1

Comment 1:
Overall, the Methods are described in sufficient detail. However, a bit more explanation of the more unusual approach of including articles published in two randomly selected months would be helpful. This is discussed further in the Discussion but a bit more insights earlier in the manuscript would be helpful.

Response
We agree that more information about the method should be given earlier in the manuscript. This is now added under “Methods” on page 6.

Comment 2:
At page 7, the text reads: "The search yielded a total number of 1031 citations. Endnote™ Version X9 was used to manage the generated research articles. After removing duplicates, 951 citations remained. All records not published for the first time in 2019 were removed, leaving 414 articles". To avoid confusion, I would suggest adding "for screening" to the sentence.
Response:
We have revised the sentence according to your advice under the section “Search Strategy” on page 7.

Comment 3
Page 10, lines 9-24: it would be helpful for the reader to learn a bit about what types of public health settings were involved in the 60 articles, what populations were engaged and what (quantitative) methods the articles report upon. That is, a bit more description of the study characteristics in the text.

Response: Thank you for this useful remark. We now added information about public health populations. This information is to be found under the section “Characteristics of Included Studies” on page 9 and in Table 1. Table 1 also includes information about diseases and related health problems that characterize populations in the clinical health studies. Furthermore, we have provided information about study designs in Table 3 and Table 4.
Comment 4
Table 1: I would suggest reversing the order of the reporting here, starting with category 1 rather than 2.
Response: The table (which now is entitled “Table 2”) is revised according to your advice.

Comment 5
The Results section is quite brief, relying too much on tables (1-3) that are not sufficiently explained and summarized in the text. Please elaborate on the findings reported in the tables to help the reader capture the key findings.

Response: In order to help the reader to capture key findings, we have now added more text in the Result section the to clarify the tables:
We have provided information about public health and clinical health populations under the section "Characteristics of Included Studies" and in Table 1.
Under the section "Health Literacy Definitions and Instruments" we have added some examples of context of studies that used definitions in Category 1, numbers of generic instruments versus context-specific and provided example of term that referred to definitions in Category 1 and Category 2"

Comment 6
Please consider arranging articles in tables 1-3 in alphabetical order.Dette ordner jeg senere 😉

Comment 7
The Discussion and Conclusion are comprehensive, and raises important reflections pertaining to results and methodological strengths and weaknesses of the work conducted. Would it be possible to raise more concrete implications of your work, that is, how do we move forward in order to overcome the identified inconsistency between between health literacy definitions and instruments in future health literacy research?

Response: Thank you for the relevant comment. We have added the following section in the manuscript under the "Discussion" on page 22:
"Clearly, our study highlights the discrepancy between definitions and measurements as well as the narrow focus of health literacy in a large amount of empirical research. An important question to raise is therefore how the findings from this specific study can guide future research strategies to overcome the identified inconsistency. In other words as the majority of health literacy researchers seem to base their research on a broad understanding of health literacy, what that can be done to facilitate increased use also of broad measurements. A first step should be to make researchers aware about the excisting mismatch in current research. Furthermore, it seems nesecerry to develop more instruments that can answer the research questions posed. Despite the high number of instruments, there still seems to be a need for questionnaires in the field of health literacy that captures other and more multidimentional different dimensions of health literacy and not only functional aspects"

Comment 8
Furthermore, the review included only quantitative measurements. What might be the value of including qualitative methods in order to create more meaningful approaches to this field?
Response;
We agree that qualitative methods can provide valuable insights into the filed. This is now discussed in the "Discussion" section, page 26.
"The current review only included only quantitative measurements. However, qualitative approaches might provide valuable and more in-depht insights in the field. For future research, it would be interesting to also explore how qualitative research link health definitions to research questions posed".

Reviewer 2
Introduction
Comment 1

The authors introduced the concept of “established health literacy theory” only in the discussion. This needs to be mentioned in the introduction section.

Response: Thank you for this remark. We have now included more information about Nutbeam`s health literacy model in the “Introduction” section on page 5.

“In 2000 Nutbeam proposed a health literacy model that is now widely cited in the health literacy literature and seen as useful in analyzing health literacy abilities required in various health situations 26

Method:
Comment 2
As mentioned by the authors, the study design is limited by the time frame. However, the number of articles is likely sufficient to answer the research question. [May need additional review by epidemiologists/ statistician who have expertise in sampling for systematic review].

Response: We have not utilized statisticial methods to look into sampling in the current review. But in order to argue better for the validiy of the method used earlier in the manuscript, (also in line with recommendation from reviewer 1), we have moved the following text from the Discsuuion to the Method section on page 6:
“Due to a high number of health literacy articles published every year, random selection was chosen in order to reflect current health literacy research. To ensure that these months were not unique in terms of the number of articles published, we performed the same search strategy using the same databases for one other randomly selected month in 2019. This search yielded nearly the same number of articles”

Comment 3
Did the investigators measure inter-rater reliability between the reviewers? If so, please report the data.
Response: We did not measure inter-rater reliability between the reviewers. We have included some text in order to better explain better how we worked to avoid selection bias under the section «Selection Criteria” on page 7:
“To avoid selection bias the researchers screened the articles for eligibility according to the criteria independently. Then they met in pairs to compare their results, resolve any conflicting opinions, and decide whether to include each article. Conflicting opinions in pairs were presented and solved by the whole team.”

Data/ Results section
Comment 4 The data presented is sparse. The lack of granularity makes it difficult to identify next steps for future research. It is insufficient to just say that 60 articles have clinical health focus and 60 in public healthsettings. The authors could categorize the articles by sectors/ subspecialty etc.

Response: Thank you for this remark. We have provided information about public health and clinical health populations under the section “Characteristics of Included Studies” and in Table 1.

Comment 5: Page 10, line 49. Section on Health literacy definitions and instruments.
In order to keep the paper consistent, please move paragraph 1 in this section to be paragraph 2 (The paragraph on health literacy definition should come before HL definition).
Response: This is now revised so that the paragraph on health literacy definitions is before the text about HL instruments.

Comment 6: Certain sections of the data are important and helps show the extent of the problem (Tables 2and 3). However, it would be even more effective to provide some granularity on which group of articles that have mutual levels or confliction levels of HL definition and HL instruments. That way, the readers can identify targets of intervention.
Response: We have now included information about type of design and type of studies (public health studies/clinical health studies) in Table 3 and Table 4.

Comment 7: Based on Appendix 2 (coding for data extraction), the investigators have collected several other variables including: perspective, country of origin, population health status, type of disease, type of HL instrument (generic/ disease-specific). Thus, the investigators should make full use of these data
Response: Thank you for this remark. We have now added more information in the “Result “section:. We have provided information about public health and clinical health populations under the section “Characteristics of Included Studies” and in Table 1.
Under the section “Health Literacy Definitions and Instruments” we have added numbers of generic instruments versus context-specific, provided an example of study context of studies that used definitions in Category 1, and provided example of terms that guided the categorization of the health literay defintions.

Discussion section:
Comment 8: Overall, the discussion section needs polishing and rethinking. Although it is well-organized, the arguments/ recommendations are often vague and abstract. Several of the paragraphs seems to create more questions than help clarify. Further details are necessary. To put it in another way, the authors often “tell”, but do not “show”. The following are some examples:
- Page 18, line 17. “As a result, there is a risk of missing important information [specifically, what is missing? What do you consider important?] about health literacy considered important in the studies’…”
- Page 18, line 22. “This concern [what specific problem? Please give more details] was previously addressed by health literacy researchers” [23]. The article in reference 23 talks about many different aspects of health literacy. I suggest choosing a specific aspect that is most relevant to your research question.

Response: We have worked with the “Discussion” section to make the message more concrete and less vague.

Comment 9
Page 18, lines 23-34. “The current study … is the first review to investigate… from the perspective of an established health literacy theory”. This is the first time the authors have mentioned this. In the introduction section, the authors have referenced Nutbeam and Sorensen. The way it is written, it read as if they are experts in the field, not that Nutbeam have established a health literacy theory. Since the audience of BMJ open are typically wide, I recommend that the authors include several sentences or one short paragraph describing more about the established health literacy theory.
Response: We have now expanded the information about Nutbeam`s model in the “Introduction” section and removed the word “established” as this expression might be to strong.
“In 2000, Nutbeam proposed a health literacy model that is now widely cited in the health literacy literature and is by many health literacy researchers seen as useful in analysing health literacy abilities required in various health situations.26 According to Nutbeam, health literacy (HL) contains three differen levels, progressing from basic skills in reading and writing (functional HL) to the ability to derive meaning from different forms of communication and apply new information to changing situations (interactive HL) and to more advanced cognitive skills which, together with social skills, can be applied to critically analyse information the ability to achieve policy and organizational changes (critical HL)”

Comment 10
Page 18, lines 50-52. “Despite the increased development of questionnaires with a broader comprehensive focus [somewhat confusing – broad vs focus – which one?], it might take more time for these instruments [which instruments? – It seems like the authors are talking about new questionnaires being develop. But they could also be talking about the new vital sign.] to become established in the health literacy research field.”

Response: The is is now revised in the Discussion (on page 20 to better clarify this message.

Comment 11
Page 19, lines 3-6. “ …, most of the focus has been on measuring individual health literacy, but all sides of the health literacy equation (including health systems and health professionals) and social processes are important to research…”. Organizational health literacy and health literacy related to health professionals are viewed in a different context. In particular, organization health literacy is measured by separate instruments. The statement above is not really relevant to the study question. I recommend removing it.
Response: We have removed the statement

Comment 12
Page 19, lines 10-12. “Definitions should be the basis of measurement. However, measurement should continually inform definitions”.
o Vague concept. Unclear what the sentences actually mean. Need to clarify further/ given more concrete examples.
Response: We have removed this statement.

Comment 13
• Page 19, lines 52-55. “An interesting finding is that, among the eight studies…. were in the context of mental health.” This is the first time this study finding is mentioned. Please include in the result section. This is a good example of some granular data that is needed in the results section
Response: This finding is now presented in the “Results” under the section «Health Literacy Definitions and Instruments” on page 10.

Editor comments:
Comment 14:
Please revise the article title to enhance readability. Eg, “Definitions and measurement of health literacy in health and medicine research: a systematic review” (or similar).

Response: Thank you for your suggestion. The title is now revised according to your advice

Comment 15:
Please revise the formatting of your abstract so that it includes the following sections: Objectives >> Design >> Data sources >> Eligibility criteria >> Data extraction and synthesis >> Results >> Conclusions. Please see the following published abstract as an example: http://bmjopen.bmj.com/content/8/3/e019438.

Response: Thank your for the example. The abstract is now revised.

Response 16: Please ensure that the abstract contains an adequate description of the search strategy and selection criteria – ie, in the “Data sources” section should list the databases searched

and date range covered by the search. The “Eligibility criteria” should include language restrictions as well as key inclusion and exclusion criteria.

Response:
The abstract now contains information about strategy and selection criteria, the databases, date range of the search, eligibility criteria and inclusion criteria.

Comment 17:Please update the Results section of the abstract to include the total number of studies included in the review and expand on phrases such as “the majority of” and “almost half” with the actual numbers of studies being referred to.
Respone: This is now revised accorindg to your advice

Cooemt 18.*Please add a statement at the end of the manuscript (before the reference list) under the heading “Data availability statement” indicating. If no additional data are available, please just state “No additional data are available.”

Response:
We have added the following text at the end of the manuscript:
“The data from the study will be made available upon request to the corresponding author”
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VERSION 2 – REVIEW

REVIEWER	Kristiansen, Maria University of Copenhagen, Department of Public Health & Center for Healthy Aging
REVIEW RETURNED	18-Nov-2021

GENERAL COMMENTS	Dear Authors, Thank you for your thorough response to my previous comments. You have addressed all suggestions well. Only very minor remaining issue relates to my comment number 6 ("Please consider arranging articles in tables 1-3 in alphabetical order). In the response letter, you've added "Dette ordner jeg senere 😉) (for editor: "I'll do this later*): I see that you have edited tables but not consistently in table 3. Please revise so that articles are listed in alphabetical order. Other than that, no further suggestions from my side.

REVIEWER	Ng, Jia Hwei Donald and Barbara Zucker School of Medicine at Hofstra/Northwell, Medicine
REVIEW RETURNED	19-Nov-2021

GENERAL COMMENTS	The authors have adequately addressed previous concerns, particularly on the results and discussion sections. There are few typo errors in the article, but that can be fixed by the copyeditor. 1. Introduction and Methods – changes made to this section are appropriate and satisfactory. 2. Results/ data – The authors have included a more detailed description of the study findings (by including an extra table and

	details on study design). This is helpful because it now delineates the scope of the problem. 3. Discussion section – The discussion section links to study findings and provides a stronger argument. The addition of the highlighted portion is concrete and will guide future researchers in the next steps. Looking forward to the published paper and future work by the authors.

VERSION 2 – AUTHOR RESPONSE

Reviewer 1 comment:

I see that you have edited tables but not consistently in table 3. Please revise so that articles are listed in alphabetical order. Other than that, no further suggestions from my side.

Response: The articles in in table 3 are now listed in alphabetical order

Reviewer: 2

The authors have adequately addressed previous concerns, particularly on the results and discussion sections. There are few typo errors in the article, but that can be fixed by the copyeditor

Response: Thank you for this comment. We have completed a thorough proofread of the text